



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Final Demo

Team 22: Solar Home Lighting

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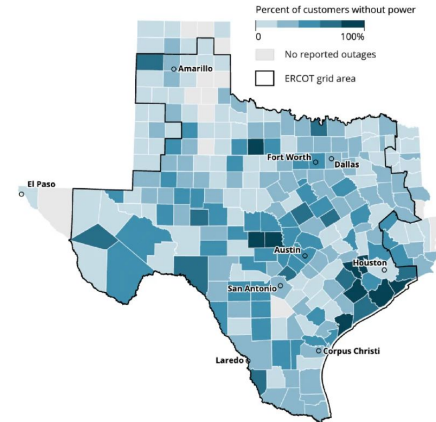
Problem Overview

Problem statement:

Current smart-home systems depend on grid power and cloud connectivity, causing lighting and monitoring to fail during outages and creating reliability and usability barriers for users.

The Solution:

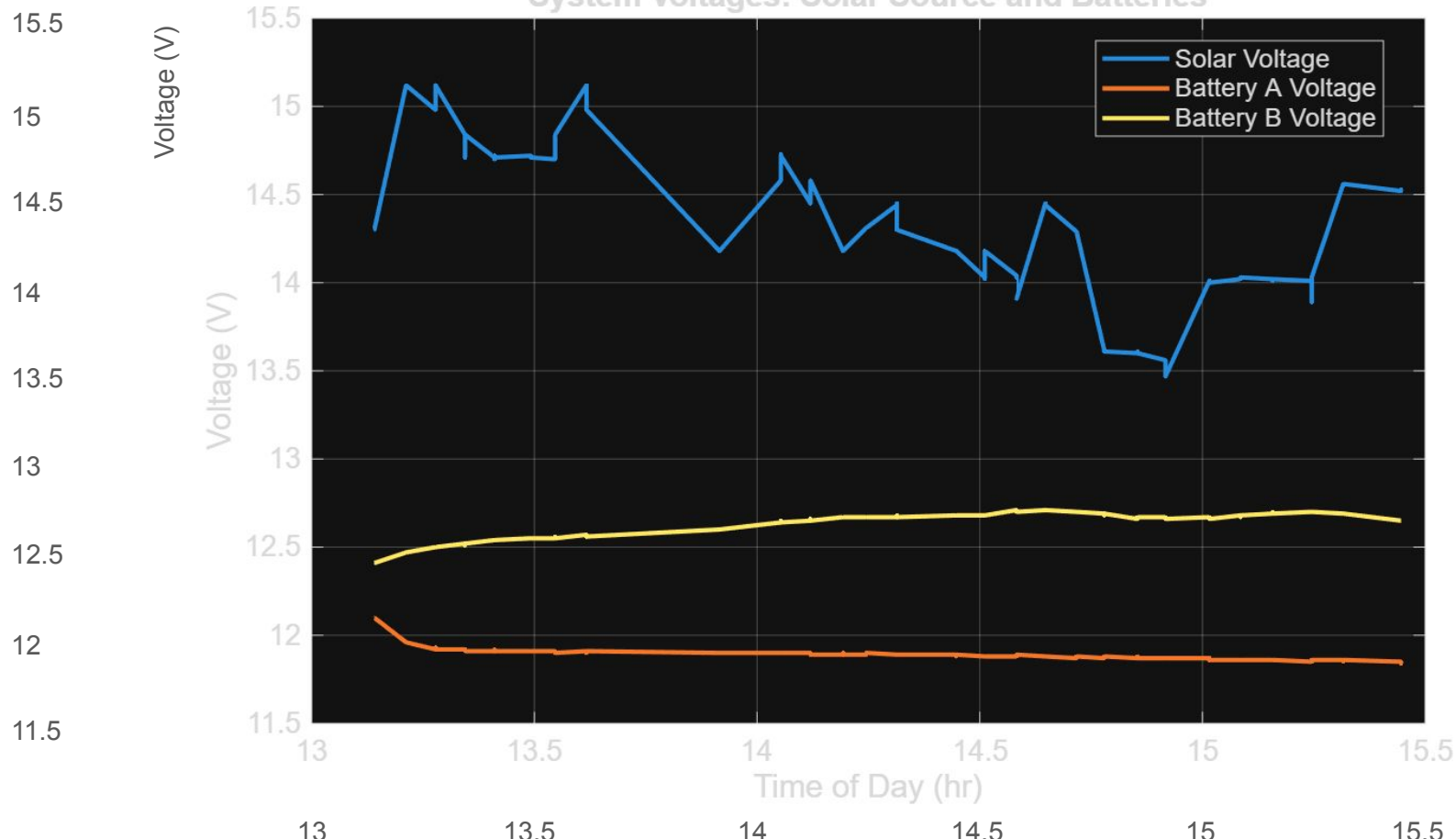
- Combine security, lighting control, and user interaction into one coordinated platform
- Offer flexible controls, with both local and remote capabilities.
- Ensures the home remains safely lit and monitored regardless of grid fluctuations.



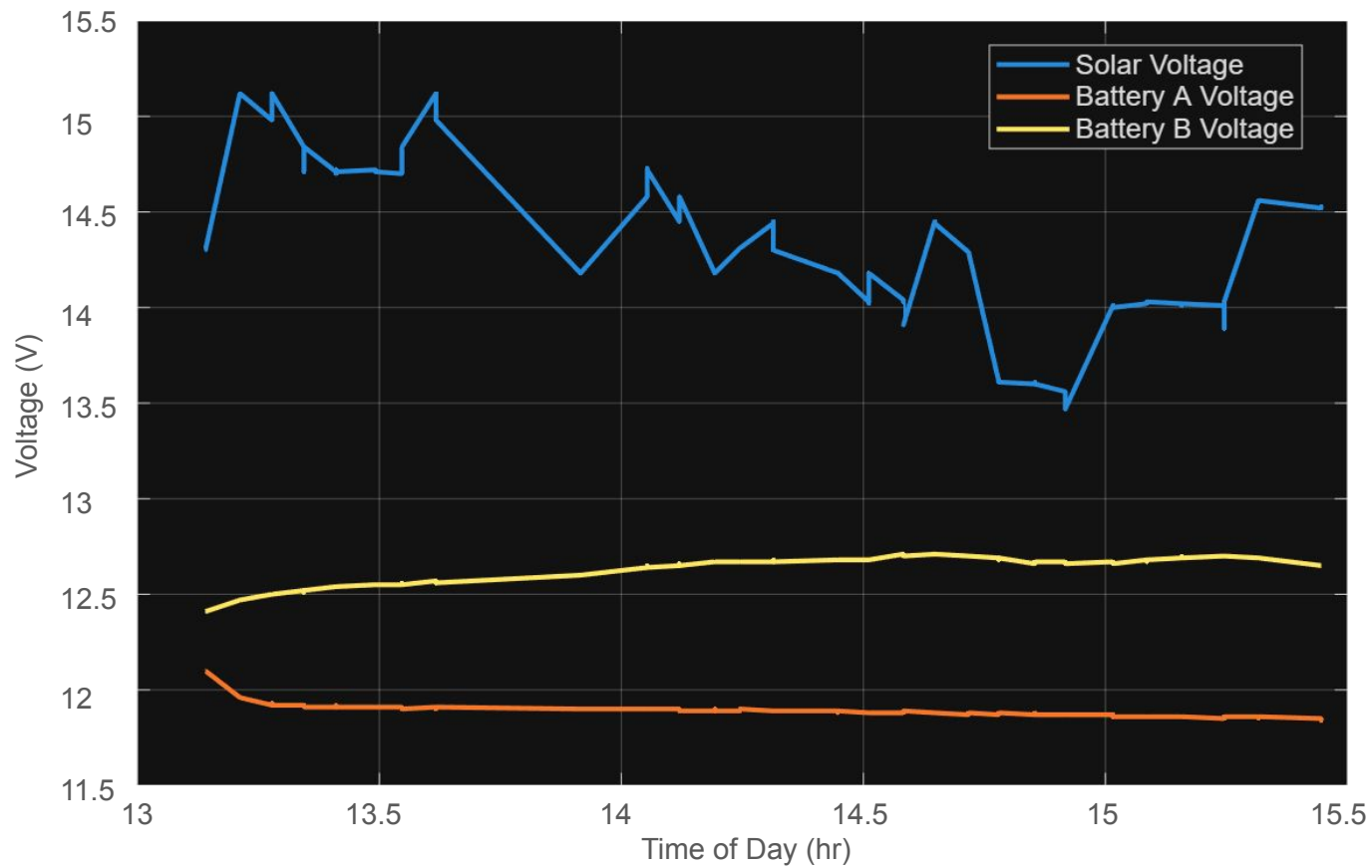
Source: The Texas Tribune and ProPublica

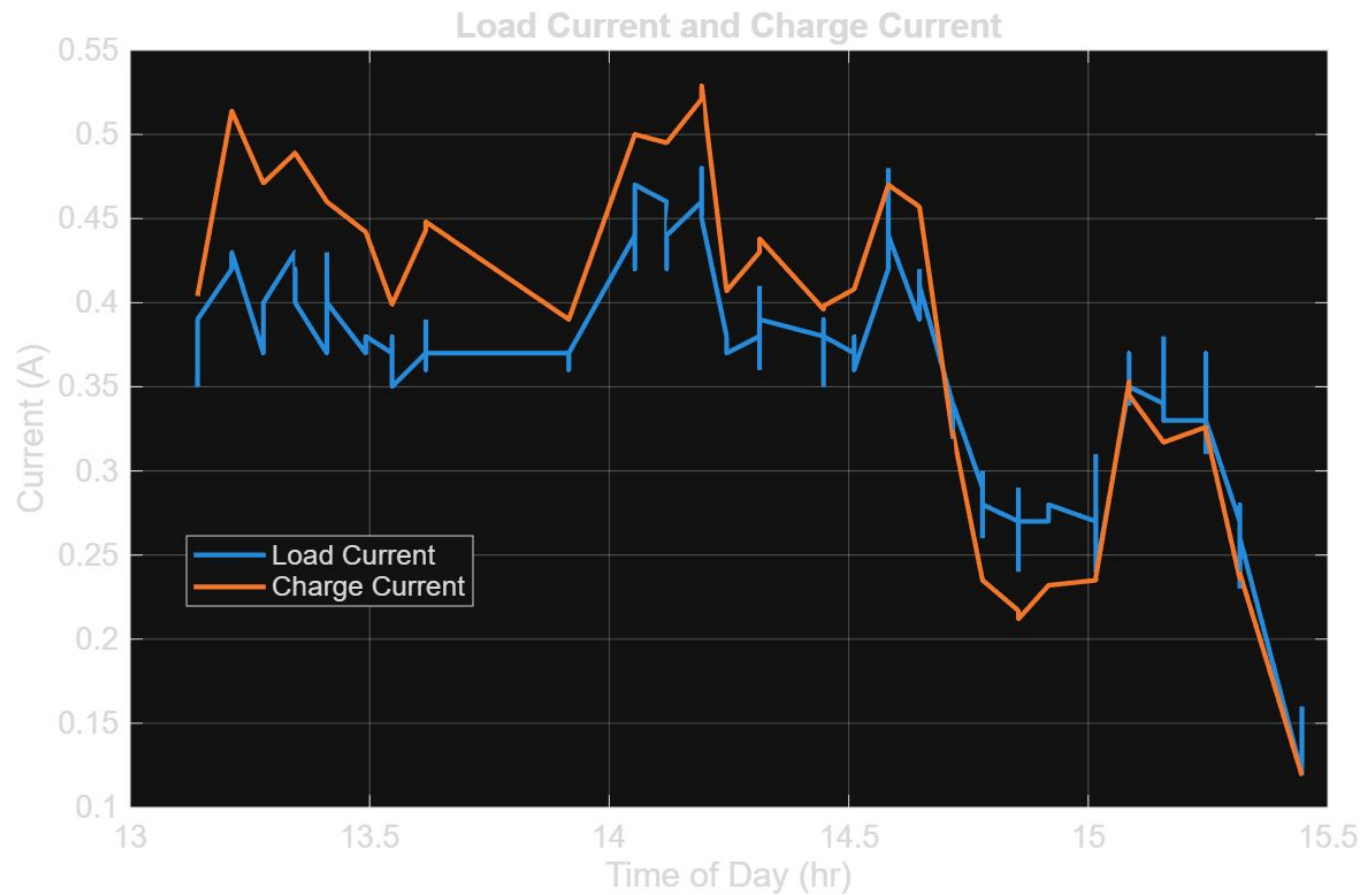
System Voltages: Solar Source and Batteries

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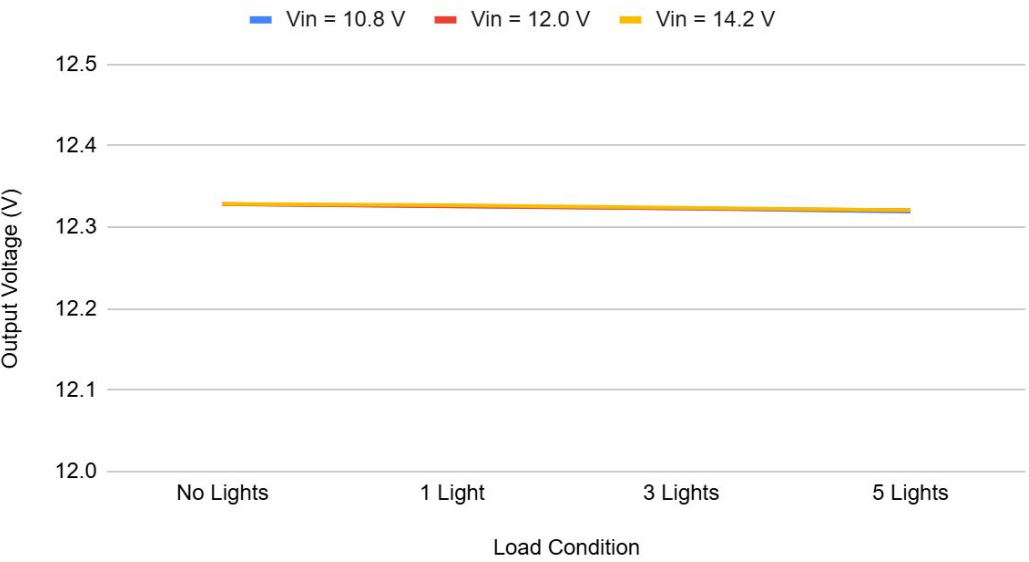
System Voltages: Solar Source and Batteries





Power Rail Stability

12 V Converter Output vs Load Condition



Metric	Value
Minimum Vout	3.333 V
Maximum Vout	3.334 V
Total Variation	0.001 V (1 mV)
Max Percent Variation	~0.03%

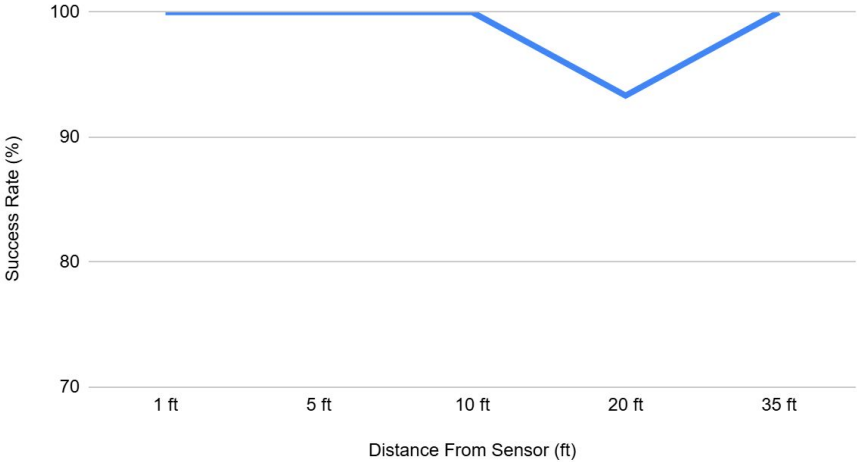
Vin (V)	Vout (V)	Error from Ideal 3.300 V (V)	Percent Error (%)
10.8	3.333	0.033	1.00%
12	3.334	0.034	1.03%
14.2	3.334	0.034	1.03%

Local Control Interface Validation Test

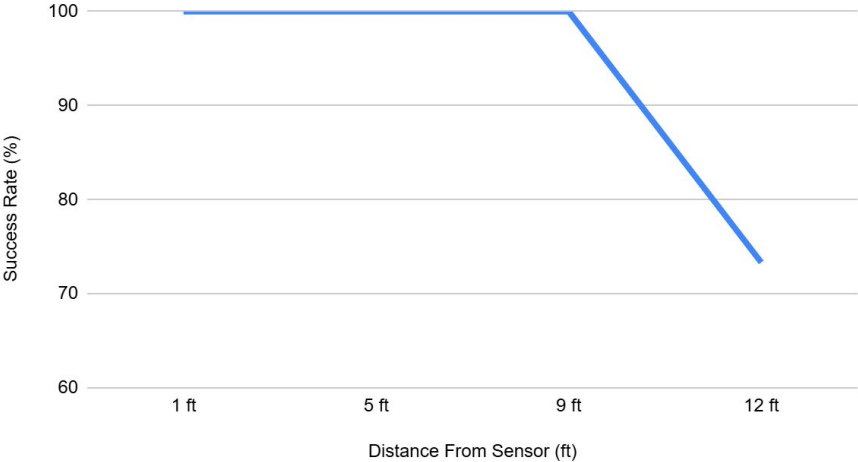
Test Scenario	Test Condition	Expected Result	Measured Result	Status
Local lighting control operation	100 manual commands issued through touchscreen (ON/OFF for Porch, Foyer, Security)	Correct lighting output activated and HMI state synchronized	100 / 100 commands executed correctly (100% success, 0 cross-trigger events)	Complete
System startup reliability	System power-cycled 30 times	HMI and lighting control initialize correctly on startup	30 / 30 successful system startups (100% success)	Complete
Continuous runtime stability	System operated continuously for 2 hours including periodic use of power-save display mode, and monitoring Battery Percentage	System remains responsive with no control failures or metric discrepancies	2 hours continuous operation with no resets or control failures; HMI recovered correctly from power-save state	Complete

Motion Sensor Reliability

Outdoor PIR Trigger Success Rate

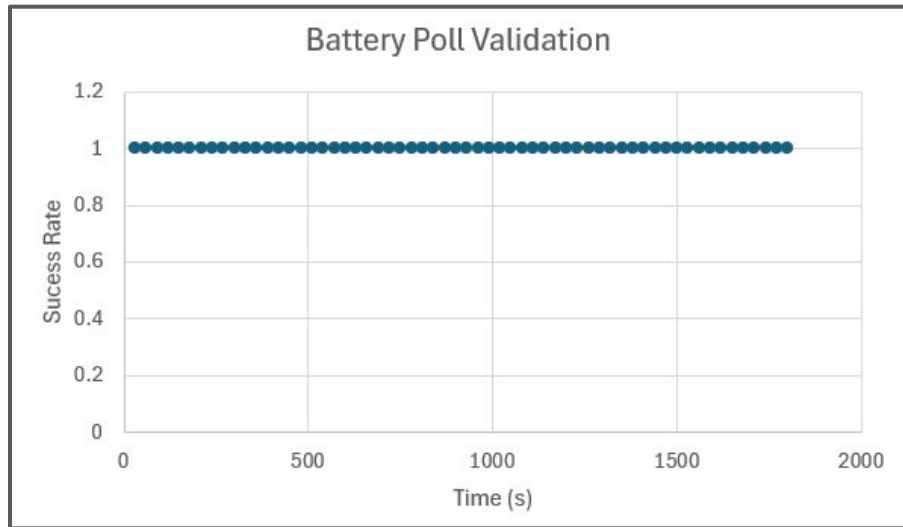
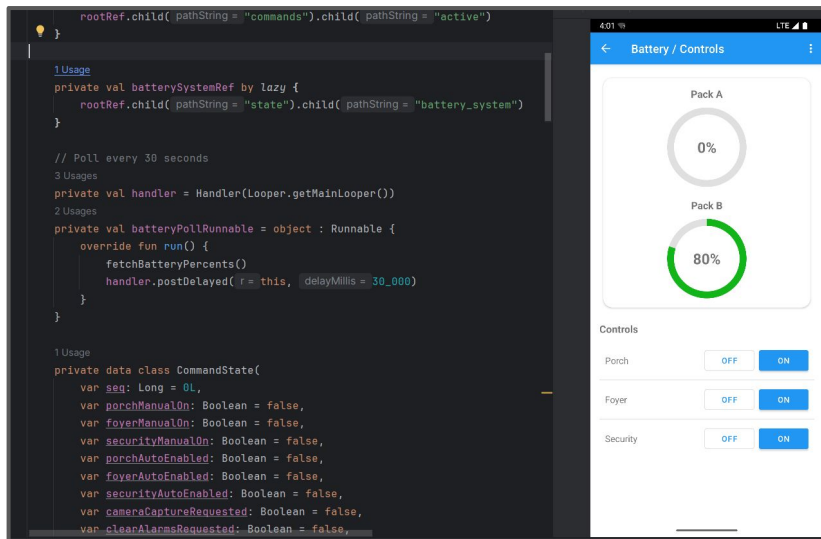


Indoor PIR Trigger Success Rate



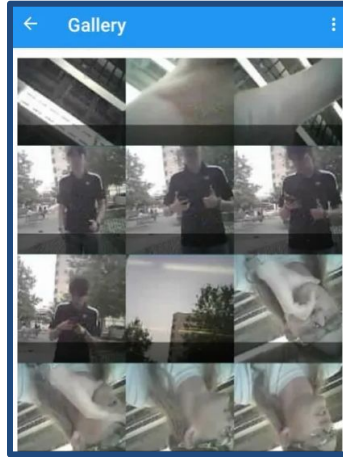
Sensor	Rated Range	Measured Performance	Result
Indoor	9 ft	100% ≤ 9 ft, 73% @ 12 ft	PASS
Outdoor	35 ft	100% ≤ 10 ft, 93% @ 20 ft, 100% @ 35 ft	PASS

Battery Poll Validation Test

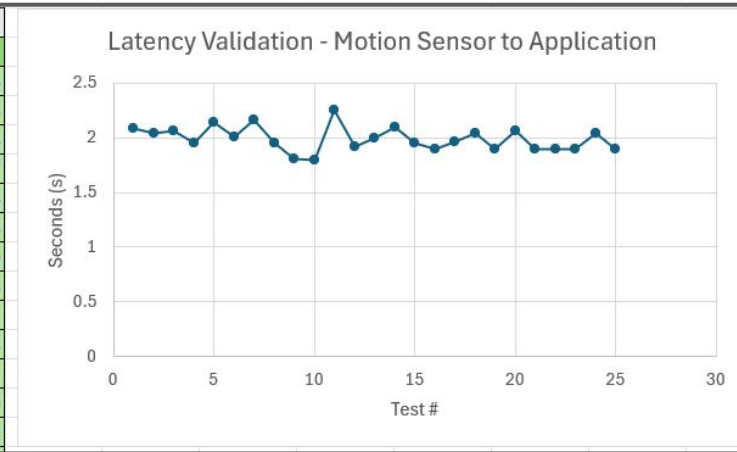


- Timer ran over 30 minutes to test Battery Poll accuracy
- Polls every 30s, not a single drop
- 100% Accuracy over 30 minute interval

Latency Validation - Motion Sensor to Application



Latency Test	
Test #	Latency
1	2.083
2	2.036
3	2.063
4	1.947
5	2.143
6	2.006
7	2.169
8	1.956
9	1.81
10	1.798
11	2.251
12	1.919
13	1.992



- 25 attempts at varying feet
- Motion Sensor -> MCU -> ESP-32 -> Database -> Application
- 1.98s average latency for whole sequence

HMI System Response Validation

HMI Latency by Zone with Target

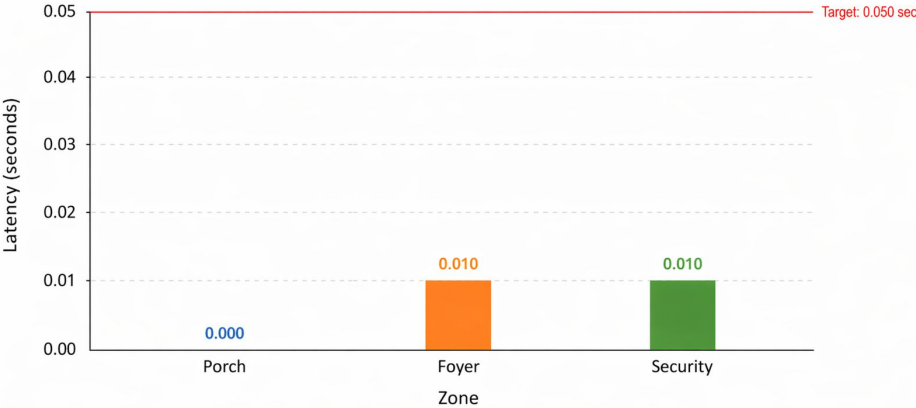


#	Validation Check	Expected Outcome	Observed Result
1	HMI command transmission over UART	Correct command string (e.g., PORCH=1) is sent to MCU	TRUE
2	MCU command processing and execution	Corresponding light state updates correctly	TRUE
3	Remote state synchronization to HMI	HMI display reflects app-controlled changes	TRUE
4	Charge controller telemetry display on HMI	Battery % and active pack are correctly displayed	TRUE

Sensor-Based End-to-End Validation

Sensor Latency by Zone with Target

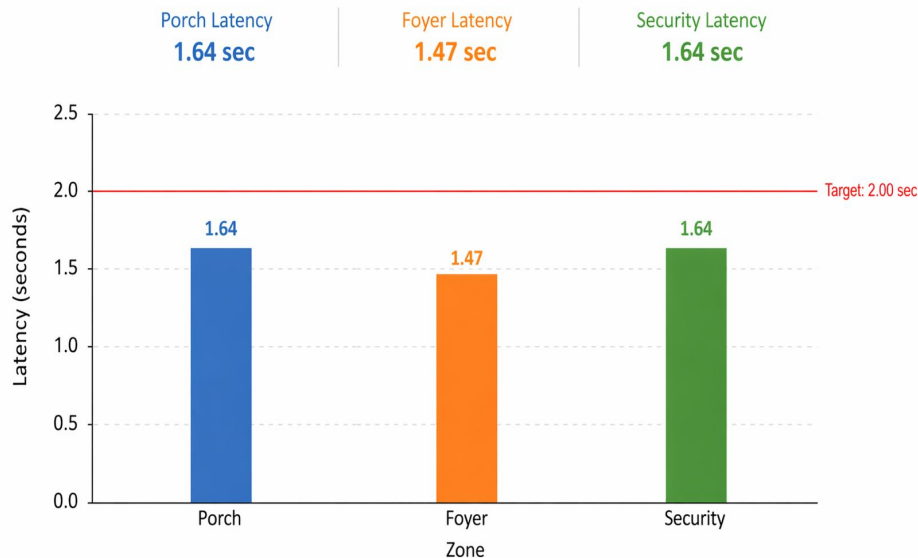
Porch Latency 0.000 sec | Foyer Latency 0.010 sec | Security Latency 0.010 sec



#	Validation Check	Expected Outcome	Observed Result
1	Sensor-triggered light activation (Auto enabled)	Corresponding light turns ON when motion/dark condition is detected	near-instant, within 0.05 s target
2	Motion-triggered notification (front door/security)	Camera triggers and notification is sent to app on motion detection	Notification received successfully (~1–2 s delay)
3	Auto mode disabled behavior	System does not respond to sensor input when Auto mode is OFF	No light activation observed
4	Sensor hold and timeout behavior	Light remains ON during motion and turns OFF after timeout (~10 s)	Correct hold and shutdown timing observed

Application End-to-End Control Validation

Remote (App) Latency by Zone with Target



#	Check	Expected	Result
1	App writes command to database	Correct control value updates in database	Complete
2	Command reaches MCU and turns on light	Light turns ON/OFF from app input	~1-2 s delay
3	App updates after light changes	App shows correct light state	~1-2 s delay
4	App shows battery data	Battery % and system data display correctly	Complete